

Maulana Azad National Institute of Technology Bhopal
Department of Mathematics, Bioinformatics and Computer Applications
Mathematics-IV (MTH-241)

Assignment-5

Due Date:

Even Sem: 2024-25

Q1. Solve the following game by using maximin (minimax) principle, whose payoff matrix is given below. Include in your answer:

- (i) strategy selection for each player,
- (ii) the value of the game to each player.

Does the game have a saddle point?

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	-2	15	-2
	A ₂	-5	-6	-4
	A ₃	-5	20	-2

Q2. Solve the following game by using maximin (minimax) principle, whose payoff matrix is given below. Include in your answer:

- i. Strategy selection for each player,
- ii. The value of the game to each player.
- iii. Does the game have a saddle point?
- iv. Is it a fair game? If not discuss the observed bias.

		Firm B				
		B ₁	B ₂	B ₃	B ₄	B ₅
Firm A	A ₁	1	7	3	4	7
	A ₂	5	6	4	5	12
	A ₃	7	2	0	3	12
	A ₄	1	11	-4	2	1

Q3. Solve the following game using the combination of Dominance Method and Maximin (Minimax) principle, whose payoff matrix is given below. Include in your answer:

- i. Strategy selection for each player,
- ii. The value of the game to each player.
- iii. Does the game have a saddle point?
- iv. Is it a fair game? If not discuss the observed bias.

Player A	Player B					
		B ₁	B ₂	B ₃	B ₄	B ₅
	A ₁	2	4	3	8	4
	A ₂	5	6	5	7	8
	A ₃	6	7	4	8	7

Q4. Two competitive manufacturers are producing a new toy under licence from a patent holder. In order to meet the demand they have the option of running the plant for 8, 16 or 24 hours a day. As the length of production increases so does the cost. One of the manufacturers, say A, has set up the matrix given below. They use the matrix to estimate the percentage of the market that they could capture and maintain the different production schedules:

		Manufacturer B		
		C ₁ : 8 hrs	C ₂ : 16 hrs	C ₃ : 24 hrs
Manufacturer A	S ₁ : 8 hrs	60%	56%	34%
	S ₂ : 16 hrs	63%	60%	55%
	S ₃ : 24 hrs	83%	72%	60%

i) At which level should each produce? And, (ii). What percentage of the market will B have?

Q5. Two computer manufacturers A and B are attempting to sell computer systems to two banks 1 and 2. Company A has 4 salesperson, company B only has 3 salesperson available. The computer companies must decide upon how many salesperson to assign to sell computer to each bank. Each bank will buy one computer system. The probability that a bank will buy from a particular computer company is directly related to the number of salesperson calling from that company, relative to the total salesperson calling. Thus, if company A assigns 3 salesperson to bank 1 and company B assigns 2 salesperson, the odds would be three to five that bank 1 would purchase company A's computer system. If none calls from either company the odds are one-half for buying either computer.

Let the payoff be the expected number of computer systems that company A sells. What strategy would company A use in allocating its salesperson? What strategy should company B use? What is the value of the game to company A? What is the meaning of the value of game in this problem?

Q6. A company is currently involved in negotiations with its union on the upcoming wage contract. The table below represents the conditional costs to the company (in lakhs). Positive signs in table represent wage increase while negative sign represents wage reduction. What are the optimal strategies for the company as well as the union? What is the game value?

		Union Strategies			
		U ₁	U ₂	U ₃	U ₄
Company Strategies	C ₁	1	7	2	3
	C ₂	6	2	7	8
	C ₃	6	1	6	6
	C ₄	4	0	5	6

Q7. Even though there are several manufacturers of scooters, two firms with brand names Janata and Praja, control their market in Western India. If both manufacturers make changes in the model in the same year, their respective market shares would remain constant. Likewise, if neither makes model changes, even then their market shares would remain constant. The payoff matrix, in terms of increased/decreased percentage of market share, under different possible conditions is given below

		Praja		
		No change	Minor change	Major change
Janata	No change	0	-4	-10
	Minor Change	3	0	5
	Major Change	8	1	0

(a) Find the value of the game.

(b) What change should Janata consider, if this information is available only to itself?

Q8. Find solution of the game theory problem using graphical method:

		Player B	
Player A		B ₁	B ₂
	A ₁	-2	0
	A ₂	3	-1
	A ₃	-3	2
	A ₄	5	-4

Q9. Find solution of the game theory problem using graphical method:

		Player B	
Player A		B ₁	B ₂
	A ₁	1	-3
	A ₂	3	5
	A ₃	-1	6
	A ₄	4	1
	A ₅	2	2
	A ₆	-5	0

Q10. Find solution of the game theory problem using linear programming method:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	3	4	2
	A ₂	1	-7	-3
	A ₃	-2	4	7
